

Applied A.I. Solutions

Data Visualization Techniques

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DATA VISUALIZATION TECHNIQUES

Data Blending

Data Combination (relationships, joins, blends)

- **Data blending is a method for combining data from multiple sources.**
- Data blending brings in additional information from a secondary data source and displays it with data from the primary data source directly in the view.
- There are several ways to combine data, each with their own strengths and weaknesses.

https://help.tableau.com/current/pro/desktop/en-us/multiple_connections.htm

Data Combination (relationships, joins, blends) – cont'd

- **Relationships** are the default method and can be used in most instances, including across tables with different levels of detail
 - Flexible and adaptable to the structure of the analysis on a sheet-by-sheet basis
 - Cannot be formed between tables from data sources published to Tableau Server or Tableau Online
 - Cannot be formed based on calculated fields

https://help.tableau.com/current/pro/desktop/en-us/multiple_connections.htm

Data Combination (relationships, joins, blends)– cont'd

- **Joins** combine tables by adding more columns of data across similar row structures
 - Can cause data loss or duplication if tables are at different levels of detail, and joined data sources must be fixed before analysis can begin

https://help.tableau.com/current/pro/desktop/en-us/multiple_connections.htm

Data Combination (relationships, joins, blends)– cont'd

- **Blends** never truly combine the data. Instead, **blends** query each data source independently, the results are aggregated to the appropriate level, then the results are presented visually together in the view.
 - Blends can handle different levels of detail and working with published data sources.
 - Blends are established individually on every sheet and can never be published, there are simply blended results from data multiple data sources in a visualization

https://help.tableau.com/current/pro/desktop/en-us/multiple_connections.htm

Data Combination (relationships, joins, blends) – cont'd

- **Blends** handle data sources at different levels of details and can be combined with relationships.
- **Relationships** have fewer technical limitations than blends and are the **recommended way of combining data when possible**.

https://help.tableau.com/current/pro/desktop/en-us/multiple_connections.htm

DATA VISUALIZATION TECHNIQUES

Data Blending in Tableau

Data Blending –

cont'd

1

Tableau - Book1

File Data Server Window Help

Connections Add

localhost MySQL

Database

gbc_superstore

Tables

- category
- city
- country
- customer
- customer_address
- order
- postal
- product
- product_order
- region
- state
- sub_category
- New Custom SQL
- New Union

product+ (GBC_Superstore)

customer_address postal

Sort fields Data source order

Field Name	Table	Remote Field Name
# Id (State)	state	id (state)
🌐 Name (State)	state	name (state)
# Region Id	state	region_id

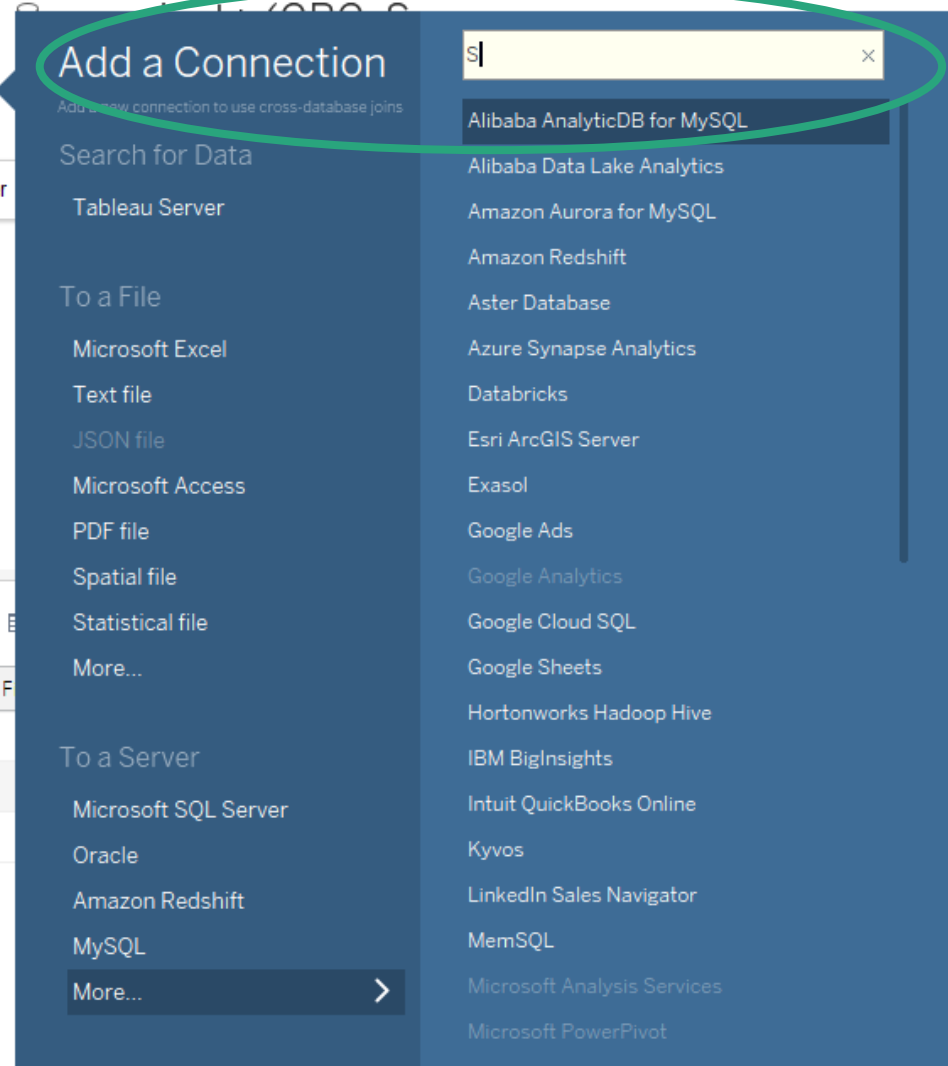
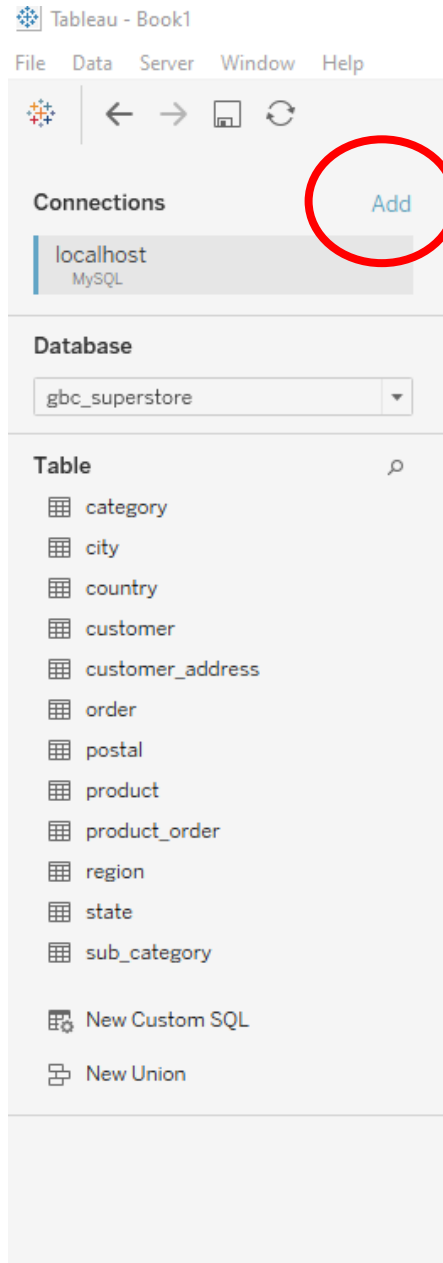
4

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Data Blending – cont'd

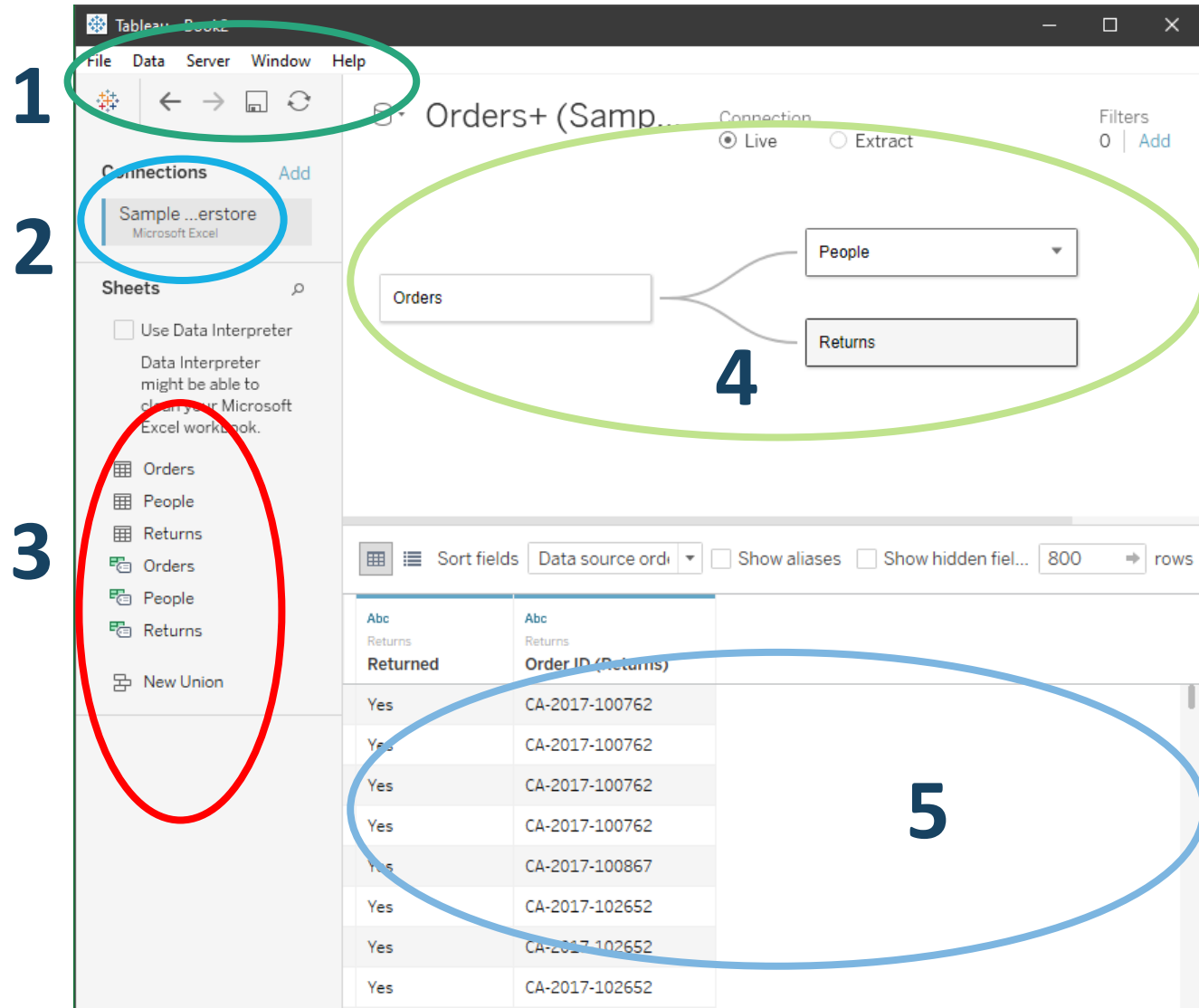
1



2



Tables and Joins – cont'd



1 Top menu bar (File, Data, Server, Window, Help)

2 Connections pane (Sample ...erstore, Microsoft Excel)

3 Sheets pane (Orders, People, Returns, Orders, People, Returns, New Union)

4 Join diagram (Orders joined to People and Returns)

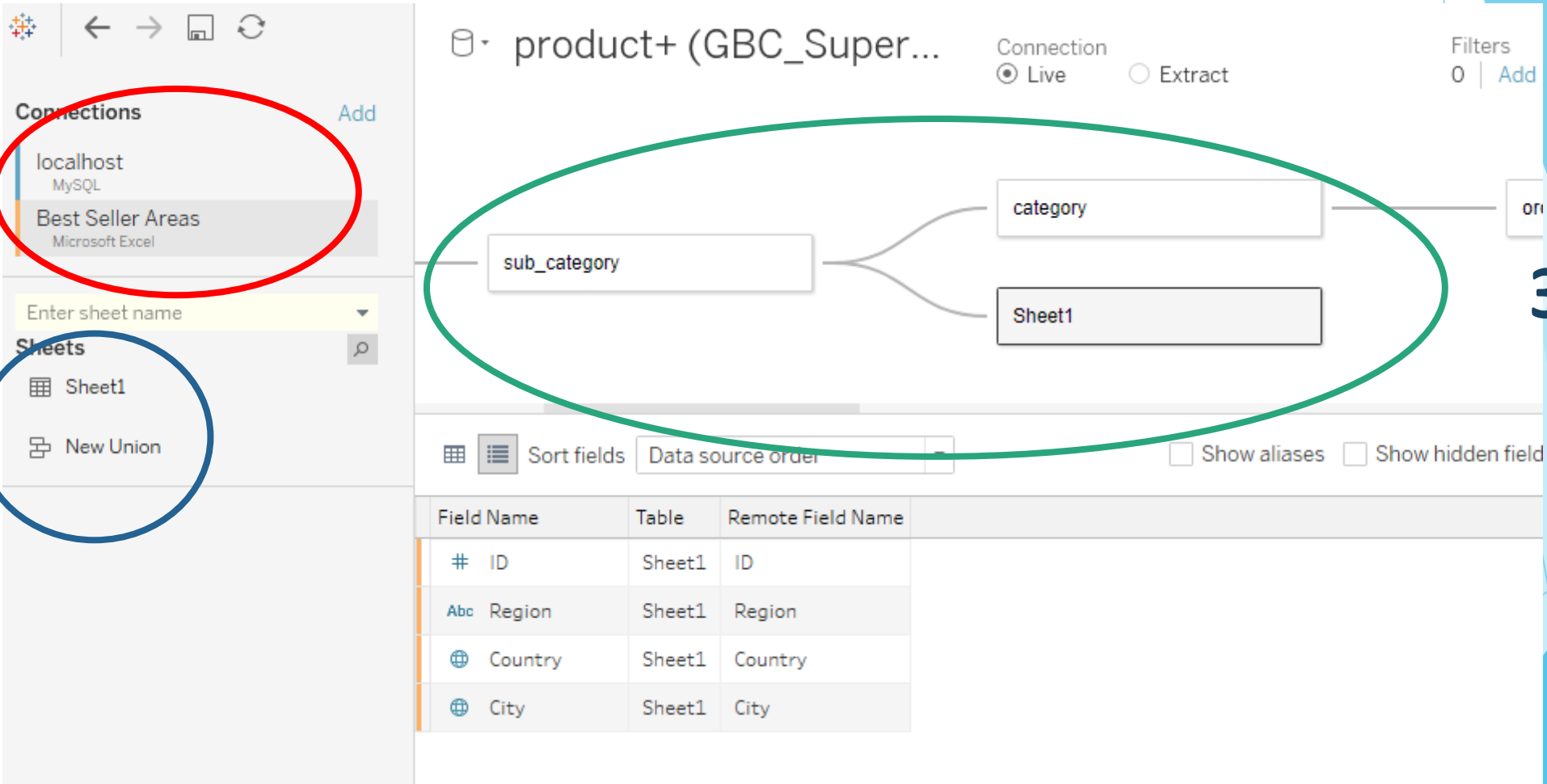
5 Data preview table

Returned	Order ID (Returns)
Yes	CA-2017-100762
Yes	CA-2017-100762
Yes	CA-2017-100762
Yes	CA-2017-100762
Yes	CA-2017-100867
Yes	CA-2017-102652
Yes	CA-2017-102652
Yes	CA-2017-102652

Tables and Joins – cont'd

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2



The screenshot shows the Tableau Desktop interface. On the left, the 'Connections' pane is open, showing 'localhost MySQL' and 'Best Seller Areas Microsoft Excel'. Below it, the 'Sheets' pane shows 'Sheet1' and 'New Union'. A red circle highlights the 'localhost MySQL' connection, and a blue circle highlights 'Sheet1'. In the center, a join diagram shows 'sub_category' connected to 'category' and 'Sheet1'. A green circle highlights this join diagram. On the right, the 'Filters' pane shows '0 | Add'. At the bottom, a table lists fields from 'Sheet1'.

Field Name	Table	Remote Field Name
# ID	Sheet1	ID
Abc Region	Sheet1	Region
🌐 Country	Sheet1	Country
🌐 City	Sheet1	City

3

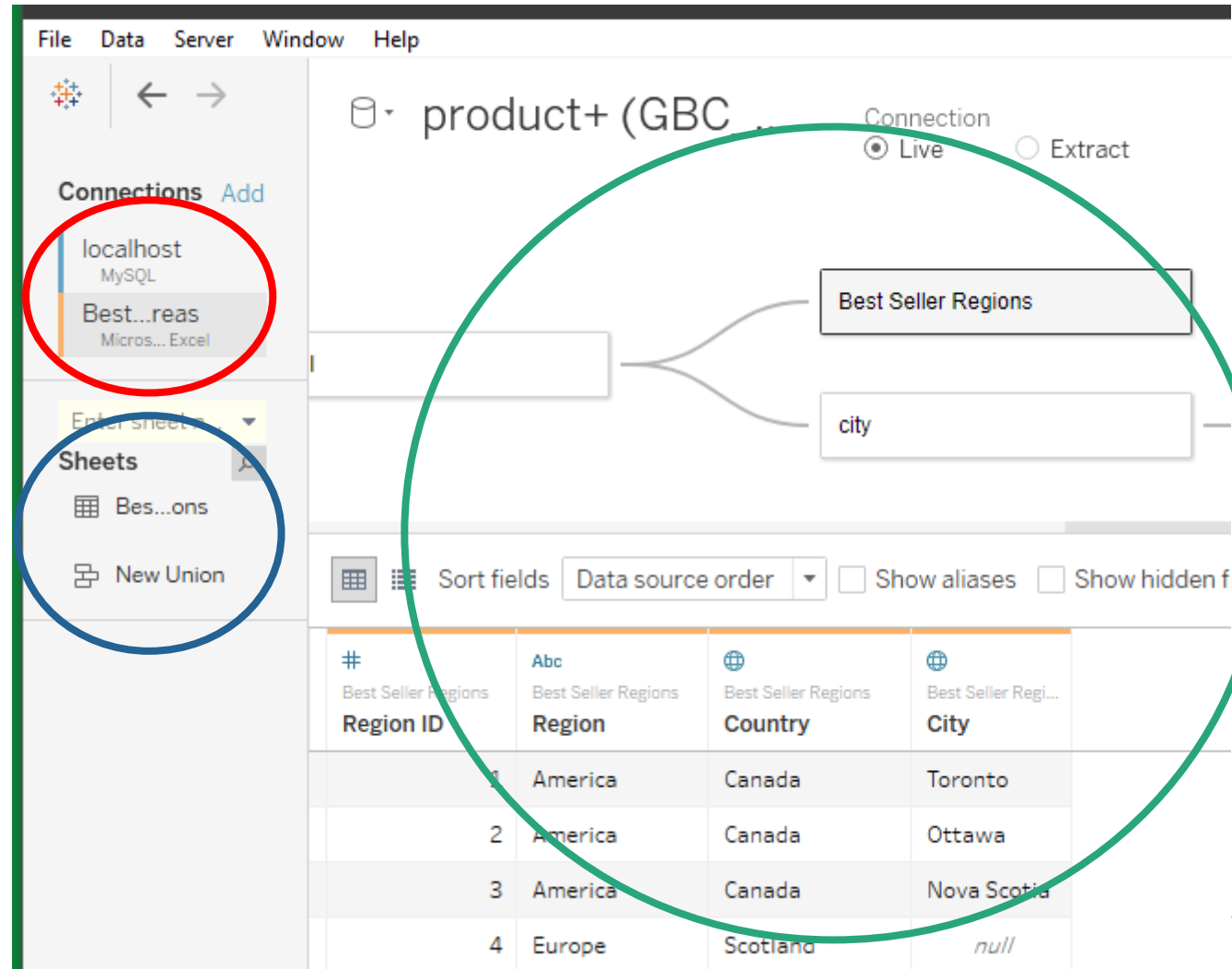
Tables and Joins

– cont'd

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3



The screenshot shows a data tool interface with a menu bar (File, Data, Server, Window, Help) and a toolbar. On the left, the 'Connections' panel lists 'localhost MySQL' and 'Best...reas Micros... Excel', with the latter circled in red. Below it, the 'Sheets' panel lists 'Bes...ons' and 'New Union', with the latter circled in blue. The main area displays a query named 'product+ (GBC' with 'Live' connection selected. A diagram shows a join between 'Best Seller Regions' and 'city'. Below the diagram, a table shows data for 'Best Seller Regions' with columns: Region ID, Region, Country, and City.

#	Best Seller Regions	Best Seller Regions	Best Seller Regions	Best Seller Regions
Region ID	Region	Country	City	
1	America	Canada	Toronto	
2	America	Canada	Ottawa	
3	America	Canada	Nova Scotia	
4	Europe	Scotland	null	

DATA VISUALIZATION TECHNIQUES

Business Intelligence tools

Tableau / MySQL Integration

TABLEAU – Data Source Selection

1

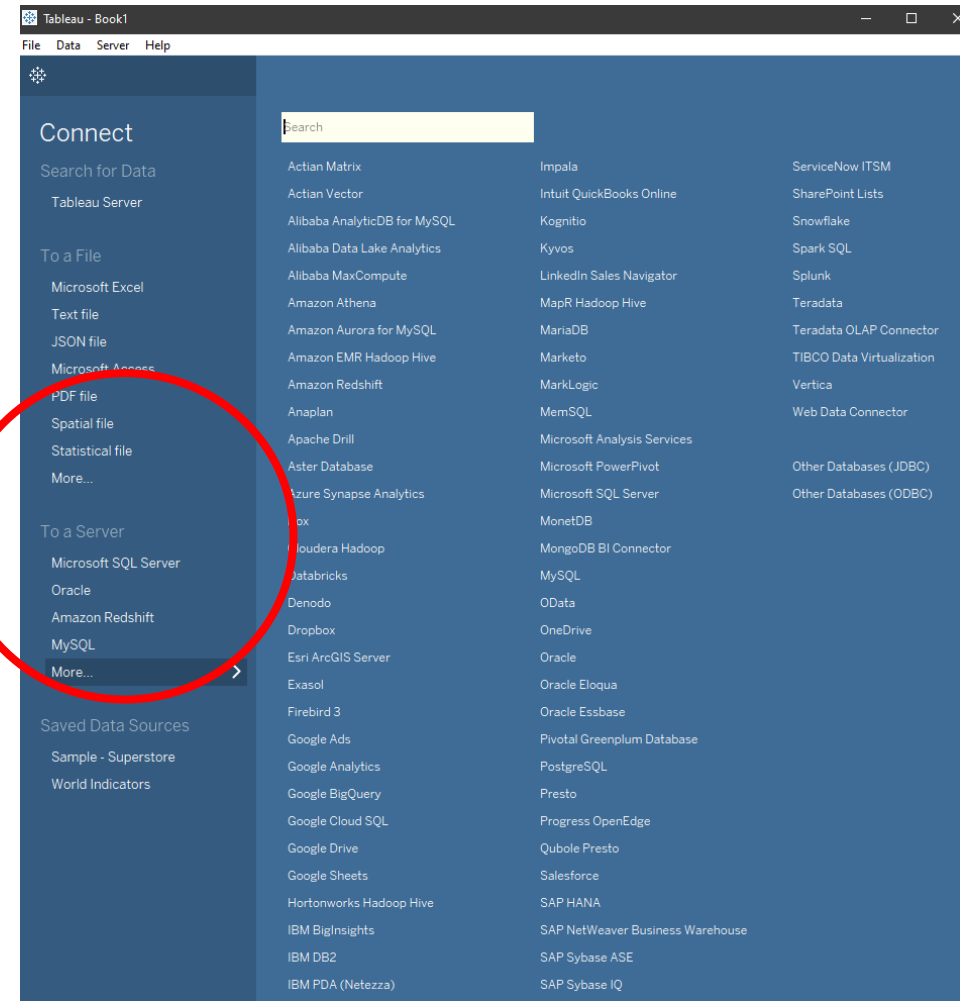
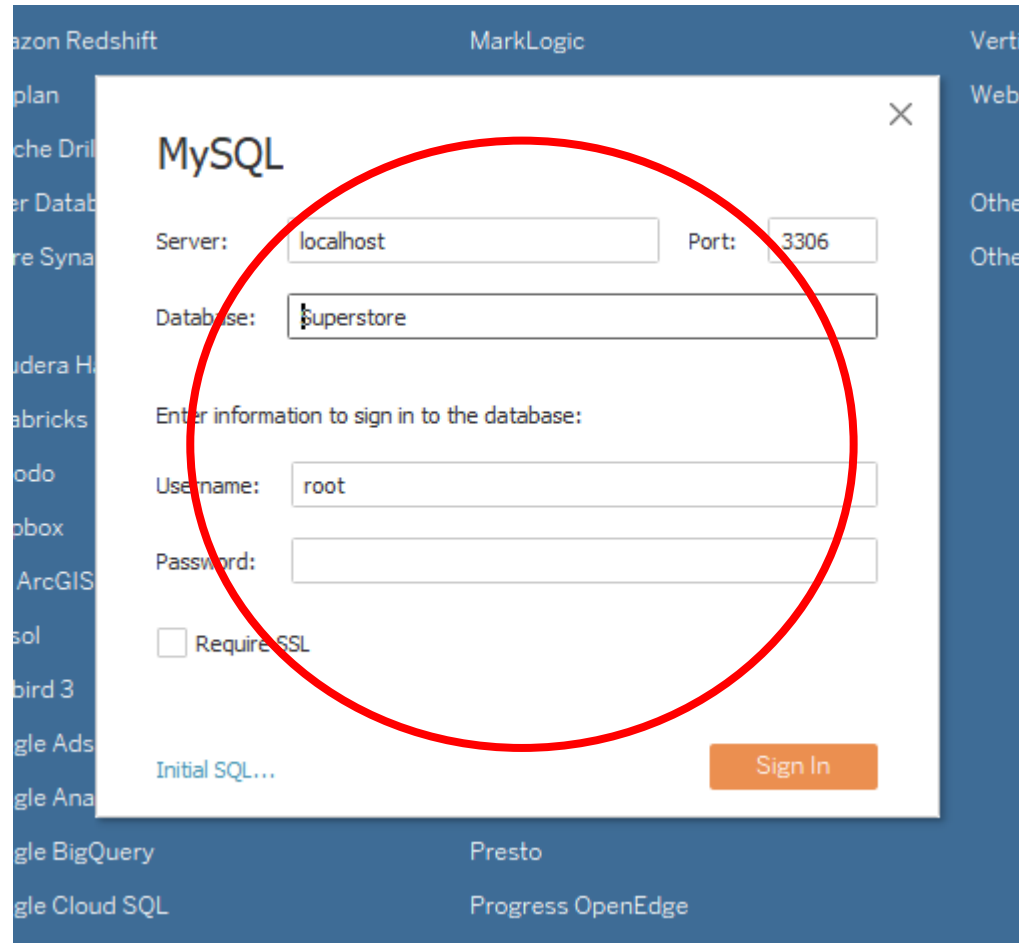


TABLEAU – MySQL Connection

1



MySQL

Server: localhost Port: 3306

Database: superstore

Enter information to sign in to the database:

Username: root

Password:

☐ Require SSL

Initial SQL...

Sign In

